Traffic Flow Prediction in Urban Areas Using Machine Learning

Machine Learning Mini-Project • UE24CS352A

PRIYANKA B M — PES1UG24AM207
PRASHANT DEVENDRA NAIK — PES1UG24AM199

1. Problem Statement

Traffic congestion is a major challenge in urban areas, where road conditions vary according to location, day, and time. Predicting congestion levels can help understand traffic patterns and support better urban traffic management.

This project implements a machine learning approach for predicting traffic congestion levels from 0 to 4 using road location, day of the week, and hour of the day. The work follows and adapts the methodology of the reference paper *Machine Learning for Traffic Flow Prediction in Urban Areas* by Yulin Huang and Genggeng Zhou. The original study investigates traffic congestion prediction using road, day, and hour as input features and a neural-network-based classifier.

2. Dataset and Data Collection

Following the methodology of the reference paper, a new traffic dataset was collected for selected routes around Central Park, New York City using Outscraper. The dataset contains traffic observations for 35 selected route queries, collected over a one-week period using the available free data collection quota.

The raw traffic data contains road/route name, distance in meters, travel duration in minutes, and timestamp. A total of **1,071 traffic records** were obtained after collection. The reference paper also used traffic data obtained through Outscraper and considered roads around Central Park, with road, day, and hour as the main predictive features.

3. Data Preprocessing and Feature Engineering

The collected data was converted into a structured format using Pandas. The timestamp was converted to New York local time, from which the following features were extracted:

• **Road (way)** – identifies the road/route.
• **Day of Week** – represented from 0 (Monday) to 6 (Sunday).
• **Hour** – represents the hour of the day from 0 to 23.

Speed (km/h) = Distance (km) / Travel Time (hours)

Unlike the reference paper's mean-and-standard-deviation-based congestion assignment, this implementation uses quantile-based labelling because the newly collected dataset has a different speed distribution. The observations were divided into five approximately equal groups.

Congestion levels: 0 – Very Low | 1 – Low | 2 – Moderate | 3 – High | 4 – Very High

4. Machine Learning Methodology

Baseline Model. The baseline uses road, day of week, and hour. The road feature was converted into a numerical representation, followed by feature scaling. A Multi-Layer Perceptron (MLP) with one hidden layer containing 10 neurons was trained using the logistic activation function. The data was divided into 80% training and 20% testing data using stratification.

Improved Model. The improved model uses one-hot encoding for road and day of week and standard scaling for hour. The neural network contains two hidden layers with 20 and 10 neurons. The logistic activation function and LBFGS optimization were used. The trained model was saved using Joblib for later prediction.

5. Results and Evaluation

Baseline MLP: 39.07% test accuracy **Improved MLP: 56.74% test accuracy**

The improved model achieved an increase of **17.67 percentage points** over the baseline. A confusion matrix was also generated to examine predictions across the five congestion classes. The improved representation of categorical features produced a better classification result than treating them as simple numerical values.

The trained model was integrated into a command-line prediction program. For a test input of **5th Ave and 86th St Transverse**, Wednesday (day 2), at 2 PM (hour 14), the system produced **Congestion Level 1 — Low**.

6. Limitations

The main limitation is the smaller dataset size and shorter collection period compared with the reference study. The dataset was collected using the available free Outscraper quota, resulting in 1,071 observations. The congestion labels were also generated using quantile-based grouping rather than exactly reproducing the reference paper's mean-and-standard-deviation thresholding. Therefore, the obtained accuracy should not be directly compared with the paper's reported validation accuracy. The observations also represent sampled time points rather than continuous traffic monitoring.

7. Conclusion and Future Work

This project demonstrates an end-to-end machine learning workflow for urban traffic congestion prediction using road, day of week, and hour as input features. The improved MLP achieved **56.74% test accuracy**, compared with **39.07%** for the baseline. The workflow covers data collection, preprocessing, congestion labelling, model training, evaluation, model saving, and prediction on new inputs.

Future work can focus on collecting a larger and more diverse traffic dataset, increasing temporal coverage, incorporating additional traffic-related features, and exploring sequence-based models such as LSTM.

Reference

Huang, Y., & Zhou, G. (2021). *Machine Learning for Traffic Flow Prediction in Urban Areas*. Stanford CS229 Project Report.